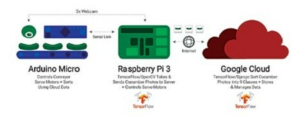
Deep learning to sort cucumbers

Inspired by AlphaGo, Makoto Koike, a former embedded systems designer in Japan, decided to build a cucumber sorter for his parents' farm, using deep learning technology. Cucumber sorting is a very difficult skill, acquired only through practice. It is difficult to hire part-time workers to help during the peak season alone, because it takes months to learn how to sort cucumbers by sight.

When Koike learnt about AlphaGo, he realised that deep learning can automatically classify images with a high degree of accuracy, using a hierarchy of numerous artificial neurons. So he decided to give automated cucumber sorting a try, using Google's open source machine learning library, TensorFlow. Although it was his first experience of deep learning, Koike got good results with TensorFlow, which does not need any knowledge about advanced Maths models or optimisation algorithms needed to implement DNNs. This gave him the confidence to get started with developing the system.

The system designed by Koike uses Raspberry Pi 3 as the main controller, to take images of the cucumbers with a camera and send it to the server. The first phase of the solution runs a small-scale neural network on TensorFlow to detect whether or not the image is of a cucumber. It then forwards the image to a larger TensorFlow neural network running on a Linux server to perform a more detailed classification. You can watch the system at work, in this video: *https://youtu.be/4HCE1P-m1l8*.

But deep learning is not without challenges, as Koike reveals in a blog post. He trained the model using around 7000 images of cucumbers sorted by his mother, but even that was insufficient. With test images, recognition accuracy was more than 95 per cent, but in real use cases, it dropped to around 70 per cent. To train the



Systems diagram of Koike's Cucumber Sorter (Image courtesy: Google Cloud Big Data and Machine Learning Blog)

system further, he would need more computing power. At present his cucumber sorter uses a typical Windows desktop PC to train the neural network model. It converts the cucumber images into 80 x 80 pixel low-resolution images, but still takes around two or three days to complete training the model with 7000 images. With this low a resolution, the system can only classify a cucumber based on its shape, length and level of distortion. It cannot recognise colour, texture, scratches and prickles. Increasing the image resolution would improve accuracy, but also increase the training time. So he looks forward to using Google's Cloud ML for this purpose, because it would give him access to a large-scale cluster for distributed training, and he can just pay for what he uses.

what these have been taught to do, so these can identify a lot more objects and sounds, and even make decisions by themselves. A deep learning system, for example, can watch video footage and notify the guard if it spots someone suspicious.

Google has been dabbling with deep learning for many years now. One of its earliest successes was a deep learning system that taught itself to identify cats by watching thousands of unlabelled, untagged images and videos. Today, we find companies ranging from Google and Facebook to IBM and Microsoft experimenting with deep learning solutions for voice recognition, real-time translation, image recognition, security solutions and so on. Most of these work over a Cloud infrastructure, which taps into the computing power tucked away in a large data centre.

As a next step, companies like Apple are trying to figure out if deep learning can be achieved with less computing power. Is it possible to implement, say, a personal assistant that works off your phone rather than rely on the Cloud? We take you through some such interesting deep learning efforts.

From photos to maps, Google uses its Brain

Brain is Google's deep learning project, and its tech is used in many of Google's products, ranging from their search engine and voice recognition to email, maps and photos. It helps your Android phone to recognise voice commands, translate foreign language street signs or notice boards into your chosen language and do much more, apart from running the search engine so efficiently.

Google also enables deep learning development through its open source deep learning software stack TensorFlow and Google Cloud Machine Learning (Cloud ML). The Cloud offering is equipped with state-of-the-art machine learning services, a customised neural network based platform and pre-trained models. The platform has powerful application programming interfaces (APIs) for speech recognition, image analysis, text analysis and dynamic translation.

Another of Google's pets in this space is DeepMind, a British company that it acquired in 2014. DeepMind made big news in 2016, when its AlphaGo program beat the global champion at a game of Go, a Chinese game that is believed to be much more complex than Chess.

Usually, AI systems try to master a game by constructing a search tree covering all possible options. This is impossible in Go—a game that is believed to have more possible combinations than the number of atoms in the universe.

AlphaGo combines an advanced tree search with DNNs. These neural networks take a description of Go board as an input and process it through 12 different network layers containing millions of neuron-like connections. A neural network called